Tarpon Cove Estuarine Habitat Restoration Project

FY 20/21

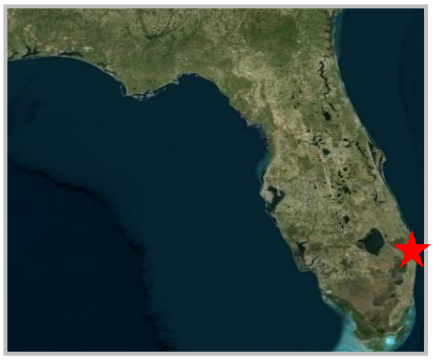
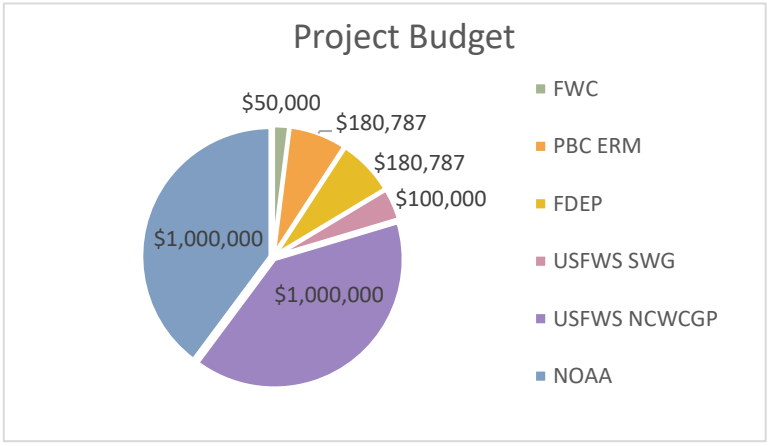


Figure 1- The Tarpon Cove Project is in the Lake Worth Lagoon in Palm Beach County in Southeast Florida



Project Partners

- Florida Fish and Wildlife Conservation Commission (FWC)
- Palm Beach County Department of Environmental Resources Management (PBC ERM)
- Florida Department of Environmental Protection (FDEP)
- National Oceanic and Atmospheric Administration (NOAA)
- US Fish and Wildlife Service (USFWS) National Coastal Wetlands Conservation Grant Program (NCWCGP)
- USFWS State Wildlife Grant Program (SWG)

Project Description

The Lake Worth Lagoon estuary in Palm Beach County (PBC) runs approximately 20 miles parallel to the southeast coastline of Florida. It is an urban estuary that has suffered extensive loss of estuarine habitats and degraded water quality due to human development activities over the past century. Approximately 81% of the lagoon's shoreline is developed with seawalls associated with private residences and businesses. Dredge activities associated with development decades ago resulted in large benthic "holes" that now impair the Lagoon's ecosystem. These "dredge holes" are typically well below grade, often anoxic, and generally do not support seagrass or wildlife. Additionally, canals that drain into the Lagoon have introduced fine, silty sediments that settle in these holes, hindering recruitment of seagrasses and diminishing water quality.

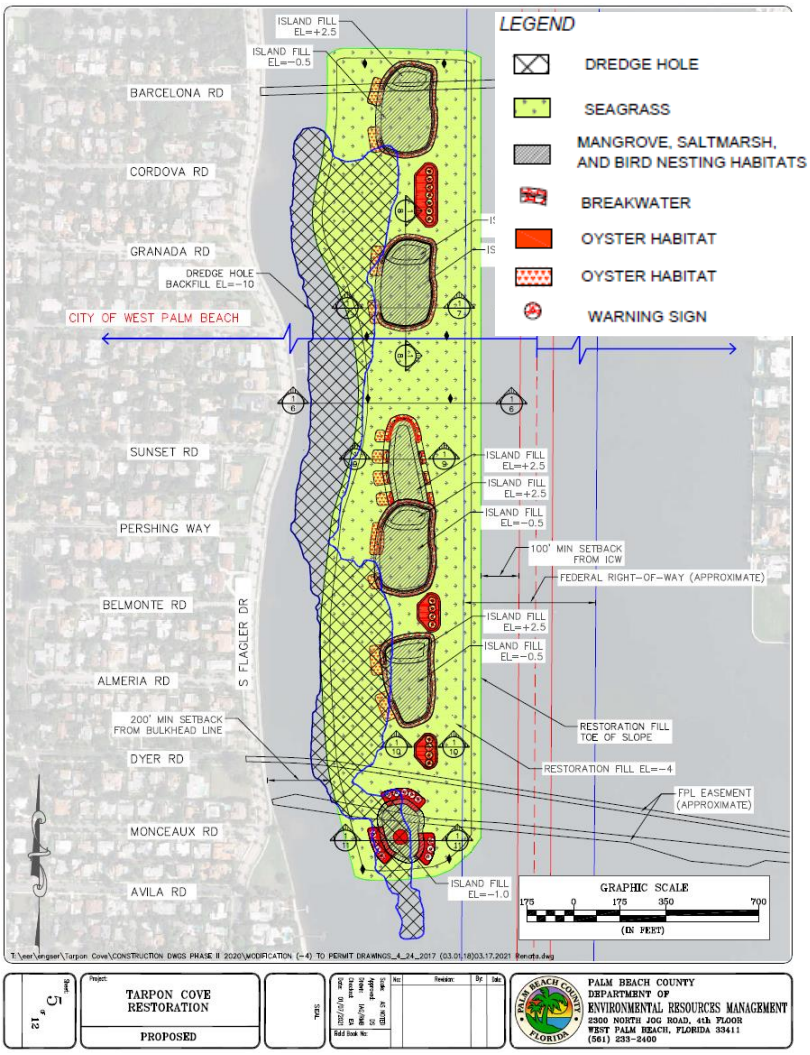


Figure 2: Conceptual drawing of the Tarpon Cove Project showing habitats that will be restored.

The Tarpon Cove project will expand restoration efforts by filling in a deep dredge hole to restore and enhance important shallow estuarine seagrass habitat through the capping of anoxic, fine-grained organic sediments (i.e., muck). Capping the muck, by filling the hole, will allow for the natural recruitment of seagrasses, including Johnson's seagrass (*Halophila johnsonii)*, a federally threatened species, to naturally recruit. Additionally, the project will result in five intertidal islands consisting of mangrove, tidal saltmarsh, intertidal sand flat, and oyster reef habitats, along with a beach- nesting bird island. These islands are expected to support over 120 species of fish, 39 species of birds, and 14 state and/or federally listed species including West Indian manatee, Johnson's sea grass, American oystercatcher and least tern. Other benefits of the project include downstream water-quality improvements, improved wildlife- and fisheries-oriented recreational opportunities, increased protection of shorelines from coastal storms and flooding, and increased carbon sequestration capabilities, the latter of which contribute to resilience to the effects of climate change. The final project, including all phases, will ultimately provide almost 40 acres of restored estuarine habitat.

Benefits

- Restoration of 2.7 acres of mangrove, tidal marsh, and oyster habitats and 34.8 acres of seagrass which will directly benefit fish and wildlife dependent on these habitats
- Creation of nesting habitat for coastal birds, including American Oystercatchers, black skimmers, and least terns
- Stabilization of adjacent hard-armored shoreline by constructing natural habitats (islands) and breakwaters to diminish wave and wind energy
- Water quality improvements as a result of reduced wave energy allowing for reduced resuspension of sediment and better conditions for recruitment of seagrasses. Increasing oysters in the area will also improve water quality.

MESS Strategic Plan Elements Supported

This proposal supports Objectives 1.1 and Strategic Actions 1.1.1 and 1.1.3 of Goal 1 of the MESS Strategic Plan.

Acreage restored by habitat type (project footprint)

Direct restoration totals 39.9 acres estuarine habitat, including:

- 34.8 acres seagrass
- 2.7 acres mangrove and saltmarsh habitats
- 2.1 acres oyster habitat
- 0.3-acre bird nesting habitat

Acreage affected by restoration activities (HUC 12): 78,706 acres



Figure 3: Construction of the northern most island. November, 2019.



Figure 4: Looking south across the northern most island of the project. Oyster habitat (riprap) can be seen in the foreground and smooth cordgrass can be seen in the background. A sign requesting that visitors not disturb nesting shorebirds can be seen mid-photo. May 2021.